

Calibrated Multi-Agent Learning Support for Project-Based Engineering Education

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Abstract

Project-based engineering education in robotics, embedded systems, and programming generates rich but fragmented learning traces. Existing generative-AI tools often return plausible final answers without diagnosing the learner's evolving knowledge state or communicating uncertainty. This proposal develops a calibrated multi-agent learning support system that integrates code, simulation records, design notes, sensor measurements, and dialogue. Specialized agents will coordinate diagnosis, evidence retrieval, feedback generation, uncertainty estimation, and instructor handoff. The system will deliver graduated support - hints, worked examples, reflection prompts, and revision suggestions - while escalating ambiguous or high-risk cases. A controlled study will test effects on learning gains, feedback uptake, transfer, and responsible reliance.

Keywords: AI in education; multi-agent systems; multimodal learner modeling; uncertainty calibration; engineering learning analytics

Background and Objective

Generative AI is increasingly available to engineering learners, yet its educational value depends on whether assistance is timely, evidence-grounded, and appropriately bounded. In project-based settings, understanding emerges through repeated cycles of coding, simulation, hardware testing, and design revision. The objective is therefore to design and evaluate an AI-supported framework that reconstructs these learning processes, provides adaptive rather than answer-only support, preserves student agency, and enables proportionate instructor oversight.

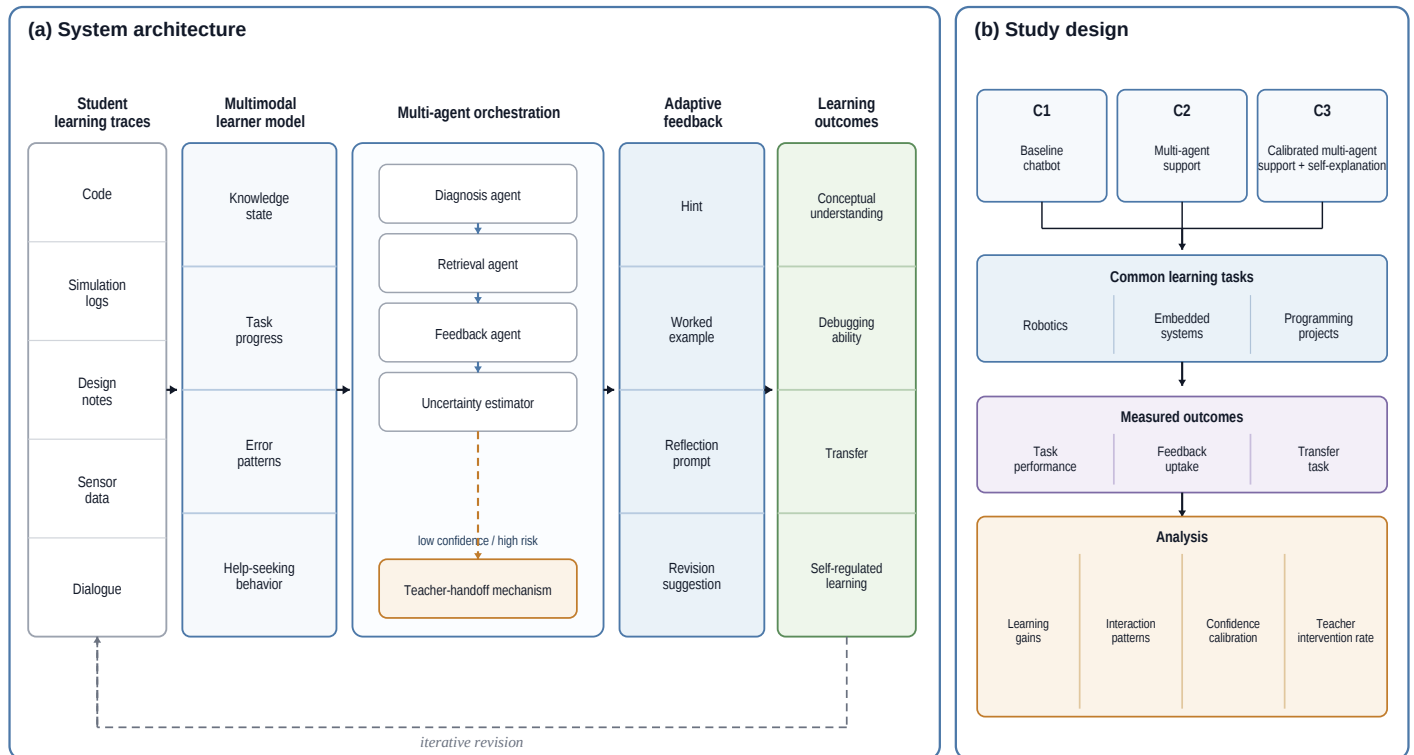


Figure 1. Proposed system architecture and comparative evaluation design.

Research Questions

RQ1. How accurately can multimodal learning traces represent knowledge state, task progress, help-seeking behavior, and recurring error patterns?

RQ2. Does coordinated multi-agent feedback improve conceptual understanding, debugging ability, and transfer compared with a baseline chatbot?

RQ3. Does uncertainty calibration combined with self-explanation improve feedback uptake while reducing overreliance and unnecessary instructor intervention?

Methodology

The study comprises three stages. First, a multimodal learner model will temporally align code edits, simulation logs, sensor signals, design artifacts, and dialogue to estimate task state and detect misconceptions. Second, an orchestration layer will coordinate diagnosis, retrieval, feedback, and uncertainty modules; responses will be grounded in course materials and student artifacts, and low-confidence or safety-critical cases will be routed to instructors. Third, participants will complete matched robotics, embedded-systems, or programming tasks under three conditions: baseline chatbot, multi-agent support, and calibrated multi-agent support with self-explanation. Pre/post performance, transfer tasks, interaction traces, feedback uptake, calibration error, and teacher-intervention rates will be analyzed using mixed-effects models and qualitative coding of revision behavior.

Evaluation and Ethical Safeguards

Primary outcomes will include normalized learning gain, debugging accuracy, transfer performance, and the proportion of feedback that leads to productive revision. Calibration will be assessed by comparing stated system confidence with feedback correctness, while interaction logs will be used to identify overreliance, disengagement, and inefficient help-seeking. The study will minimize data collection, pseudonymize learning traces, obtain informed consent, and prevent autonomous grading or consequential decisions. Instructor review will remain mandatory for safety-critical, sensitive, or persistently uncertain cases.

Expected Contributions

The proposed research will contribute (1) a multimodal learner-modeling approach for authentic engineering projects; (2) a calibrated multi-agent architecture integrating retrieval-grounded feedback, uncertainty-aware escalation, and self-explanation; and (3) empirical evidence about when AI support strengthens learning and transfer without displacing productive struggle or instructor judgment.